

This document provides a comprehensive, step-by-step template for building a BrainLift. Each section includes a detailed explanation of its purpose and structure, designed to guide you from foundational knowledge to advanced, original insights. Crucially, a BrainLift is designed to serve as structured context for AI conversations, allowing you to easily share your curated knowledge with any AI model.

Owners

- Anshul Mago

Purpose

Build a high-stakes exam prep tool that gives only evidence-backed readiness estimates, with uncertainty and topic coverage built in, and recommends the analytically best next study action based on learning science rather than raw percent correct

Most learning apps confuse percent correct with readiness; We treat learning as a calibrated decision problem and only makes claims when the evidence supports them

Explanation: Most study tools treat learning as a tally of correct versus incorrect responses, but that collapses several different constructs into one noisy score. Your product treats student learning as a decision problem under uncertainty: it separately models memory, transfer, and readiness; adjusts for missing topic coverage; and only recommends a next action when there is enough evidence to justify it. The app's novelty is that it operationalizes findings from retrieval practice, metacognitive calibration, and self-regulated learning into a system that tells the student not just how they are doing, but what they should do next and how sure the system is about that recommendation.

In Scope

-Learning science and exam modeling: Capture evidence-based study principles and detailed exam outlines/scales to design, test, and interpret study features and readiness metrics for high-stakes exams.

-Anki, FSRs, and shared-engine architecture: Map Anki's Rust/Python/protobuf/mobile stack and define concrete engine changes (queues, mastery queries, sync rules) that power both desktop and phone apps on one shared backend.

-AI engineering for study tools (RAG, evals (harness), safety): Design source-grounded RAG pipelines, evals, baselines, and guardrails so every AI output is traceable, tested on held-out data, and aligned with strict abstention and honesty rules.

-Engineering practices and research workflows: Systematize build, packaging, benchmarking, crash/offline/sync tests, and UI patterns so experiments and products are reproducible, calibrated, and expose memory/performance/readiness transparently.

Out of Scope

This clarifies what you will **not** be focusing on. Setting clear boundaries is crucial for preventing scope creep and maintaining focus.

Using AI to synthesize, summarize, or create new knowledge (e.g., insights).

Relying on AI to replace the process of information passing through your own brain.

DOK 4: Spiky Points of View (SPOVs)

This is the pinnacle of the BrainLift (Depth of Knowledge Level 4). A Spiky POV is not just an opinion; it is a well-reasoned, actionable, and often contrarian argument that you have developed through the synthesis of multiple insights. It is “spiky” because it takes a strong, defensible stance that is unique and not easily generated by a standard AI model.

How it Breaks Down:

A Spiky POV is forged by combining and overlapping multiple **DOK 3 Insights** from different domains or sources. It adapts an idea from one context to solve a problem or offer a new perspective in another. The goal is to create a new mental model or a new “rule” for thinking about your topic.

- **Spiky POV 1:** Current Learning Apps and Preparatory Courses suck at measuring true readiness; they reduce preparation to isolated quiz scores (can be a result of high variance), whereas high-stakes exams demand a continuously interwoven model that updates memory, transfer, coverage, uncertainty, etc., in real-time.

- **Elaboration:** Current MCAT preparation tools treat study progress as a tally of correct answers; a low-utility metric that conflates recognition with long-term mastery. As evidenced by Dunlosky's high-utility research and FSRS-6 benchmarking, true readiness is not a percentage, but a probabilistic state. A sophisticated system must move beyond static scoring to treat learning as a decision-under-uncertainty problem. By integrating FSRS-calibrated memory models with passage-based performance indicators, interleaving strategies, and AAMC-aligned confidence bands, we shift the focus from "how much have you done?" to "how sure are we that you are ready?" This creates a continuous feedback loop where the system only recommends a next best action when evidence reaches a threshold, thereby preventing the "illusion of competence" inherent in traditional, blocked-study applications. We should always be able to tell the next best action for studying based on a student's current preparation situation.

Competitive Analysis

DOK 1: Facts (The Competitive Landscape)

All factual data points regarding competitor methodologies and student performance correlations.

How Each Competitor Measures Readiness

The MCAT prep industry largely relies on a uniform readiness proxy: percent correct on practice questions or full-length exams, mapped to a scaled score with proprietary formulas (MCAT prep company marketing pages; MCAT subreddit back-testing threads).

- **Princeton Review:** Score reports rely on raw percent-correct on proprietary full-length exams, which are then mapped to “predicted MCAT scores.” Independent regressions on student-reported data show that Princeton Review FL scores explain only a small fraction of the variance in real MCAT outcomes; the R-squared values in these community analyses are markedly lower than those observed for AAMC practice exams (student back-testing spreadsheets compiled in r/Mcat; “AAMC vs. Princeton Review FL” threads). In practice, this makes their FL scores weak and noisy readiness signals.

- **Kaplan:** Kaplan uses QBank accuracy and full-length performance mapped to predicted scores. Analyses of student score trajectories indicate modest predictive value—R-squared values in the low-0.3 range in community regressions—and Kaplan’s own materials do not describe any time-aware decay model; performance is treated as a static function of accuracy, regardless of how long ago the practice occurred (Kaplan MCAT product descriptions; community regression posts summarizing “Kaplan FL vs. real MCAT”).

- **Blueprint:** Blueprint provides one of the most sophisticated UIs, breaking results down by subject, AAMC reasoning skills, question difficulty, and answer-switching behavior (Blueprint Help Center, “Proven Strategies from Learning Science”). Blueprint has published that its FL exams correlate moderately with real MCAT scores (correlations in the ~0.7 range), which is better than many other third-party exams but still weaker than AAMC FL1/FL2 (Blueprint blog posts on “How predictive are our full-length exams?”; MCAT subreddit back-testing spreadsheets). However, their readiness analytics still do not explicitly model time-decay or distinguish between simple recall and transfer to novel passages.

- **UWorld:** UWorld is widely regarded in the MCAT community as having the most representative question style and difficulty, closely mirroring AAMC passages (r/Mcat threads such as “Is UWorld representative of the real MCAT?”). Nevertheless, its readiness reporting is still built on subject-based percent correct in the QBank. This fails to account for the well-documented “illusion of competence” that arises when students repeat similar questions and see rising percentages without corresponding gains on unseen, exam-style items (Karpicke & Roediger, 2008, *Journal of Memory and Language*, pp. 181–185; Bjork & Bjork, 2011, “Making Things Hard on Yourself, But in a Good Way”).

DOK 2: Summary (The Critical Gaps)

Synthesized findings regarding why current metrics fail to predict actual test-day success.

The Four Critical Gaps

The structural failure of current prep tools is rooted in the “illusion of competence”, where students mistake recognition fluency (re-seeing familiar material) for genuine retrieval strength

and transfer ability (Karpicke & Roediger, 2008, Journal of Memory and Language; Bjork & Bjork, 2011, APS Observer).

1. Temporal Blindness

Dashboards provide a static snapshot of past accuracy and do not account for memory decay as established by forgetting-curve research going back to Ebbinghaus and extended by modern work on storage vs. retrieval strength (Bjork & Bjork, 2011). A student who studied Biochemistry intensively two months ago will still appear “strong” in most tools, even though retrieval strength has dropped substantially, because none of the major prep platforms integrate a time-aware memory model like FSRS into their readiness calculation (MCAT prep product docs; Anki/FSRS notes on decay and review scheduling).

2. Illusion of Competence

Karpicke and Roediger (2008) show that students who rely on re-reading and recognition often predict high performance but then score dramatically lower on delayed tests of the same material (Karpicke & Roediger, 2008, JML, pp. 181–185). When prep tools emphasize repeated QBank questions and percent-correct, they reproduce this pattern: students see rising accuracy on familiar items and infer readiness, but their performance on novel, passage-based questions—the actual MCAT format—remains much lower (Taylor & Rohrer, 2010, Applied Cognitive Psychology, on discrimination errors; MCAT subreddit “I felt ready but bombed the real exam” threads).

3. Coverage Blindspot

Current tools do not penalize readiness scores when students ignore entire high-weight MCAT sections; dashboards simply show those areas as “not started” or blank. A student can study Biology and Chemistry heavily while barely touching CARS or PSBB and still receive a reassuring overall score estimate. In contrast, Derivita’s K-12 “Readiness Score” explicitly scales down readiness when only a small fraction of a blueprint’s standards have been assessed, guarding against inflated estimates based on narrow coverage (Derivita support article “Understanding the Readiness Score,” pp. 1–3). MCAT prep platforms generally lack this kind of coverage-adjusted readiness logic.

4. No Validated Uncertainty

The AAMC reports official MCAT scores with confidence bands (± 1 for section scores, ± 2 for total), explicitly acknowledging measurement error and day-to-day variability (AAMC “Understanding Your MCAT Score,” p. 2). Prep companies, by contrast, present single-number estimates with no intervals, even though community back-testing shows that third-party FL scores can be off by 5–10 points in either direction for the same student (MCAT subreddit back-testing spreadsheets; “Add 10 points to Princeton” lore). This creates a calibration error where students treat noisy, uncalibrated practice scores as precise, high-confidence predictions, when in reality the uncertainty is large and mostly unreported.

Experts

This section is a curated list of the leading thinkers, researchers, and practitioners in your domain. Following their work is essential for building the foundational knowledge needed for your BrainLift.

- **John Dunlosky**

- **Who:** Professor of Psychology at Kent State University.
- **Focus:** Meta-analysis of learning techniques; distinguishing high-utility (spaced practice, testing) vs. low-utility (highlighting) strategies.
- **Why Follow:** His framework is the bedrock for the BrainLift's argument that current MCAT prep apps are failing; he provides the evidence-based evaluation criteria for what makes a study technique "high-utility."
- **Where:** [Kent State Profile](#)

- **Robert A. Bjork**

- **Who:** Distinguished Research Professor, UCLA.
- **Focus:** Cognitive psychology of human memory; developed theories of "desirable difficulties," "retrieval strength" vs. "storage strength," and the benefits of interleaving.
- **Why Follow:** His work is foundational for the Spiky POV on interleaving MCAT topics. His research on test-potentiated learning explains why blocked studying creates an "illusion of competence."
- **Where:** [Bjork Learning and Forgetting Lab](#)

- **Doug Rohrer**

- **Who:** Professor of Psychology, University of South Florida.
- **Focus:** Interleaved practice in mathematics and complex problem-solving.
- **Why Follow:** His research provides the closest analog to MCAT reasoning, demonstrating that interleaved practice significantly reduces "discrimination errors" (confusing which problem-solving procedure to apply).
- **Where:** [USF Faculty Profile](#)

- **The Learning Scientists (Yana Weinstein & Megan Sumeracki)**

- **Who:** Cognitive scientists and authors.
- **Focus:** Bridging the gap between cognitive psychology research and practical application for students and educators.
- **Why Follow:** They excel at translating complex learning science into actionable, day-to-day study habits, which is essential for defining the user-facing readiness metrics in the BrainLift.
- **Where:** [learningscientists.org](#)

DOK 3: Insights

Insights are the building blocks of your Spiky POVs (Depth of Knowledge Level 3). An insight is an original conclusion, connection, or idea that you have generated after processing information

from your sources. It goes beyond simple summary (DOK 2) to represent your own critical thinking.

How it's Structured and Supports SPOVs:

Each insight should be a complete thought, derived from one or more sources in your Knowledge Tree. You should group insights thematically. Later, you will synthesize these individual insights to construct your more complex DOK 4 Spiky POVs. An insight is the bridge between the raw information you've gathered and the novel arguments you will make.

- **Insight 1:** Retrieval as a generative process, not a static record. Memory is reconstructive, not a 'storehouse' where knowledge is simply recorded. Therefore, the act of retrieval is the primary mechanism of learning. AI tutors that provide frictionless access to answers often create an 'illusion of competence' and degrade long-term retention by bypassing the generative effort required to build storage strength.
- **Insight 2:** Interleaved, gradual complexity. Blocking study phases (learning then testing) is inferior to 'test-potentiated learning,' where practice opportunities are interleaved directly into the learning process. For high-stakes, multi-year exams, simple flashcards are insufficient. Effective study systems must evolve from basic fact recall to complex, gradual problem sets that force learners to bridge foundational facts with the application-heavy demands of the exam.
- **Insight 3:** Context-sensitive scaffolding. Concrete examples and visual diagrams are only effective when they are structurally intertwined with the underlying abstract concepts. If visuals are purely decorative, they waste cognitive resources. If they are inconsistent with the abstract rules, they mislead the learner. Effective scaffolding requires that every visual or concrete example directly serves the abstract model being taught, preventing flawed mental models.

MCAT-aligned DOK3 – Experimental designs, formulas, and evaluation plans

1. MCAT memory model calibration (FSRS + your modifications)

- **Objective**
 - Show that your memory model (possibly FSRS-based) is calibrated on MCAT decks, with performance comparable to FSRS-6 benchmarks and reported via log loss, RMSE(bins), and Brier score on held-out BBFL/CPBS/PSBB/CARS reviews.
- **Design**
 - Data: MCAT-specific decks with FSRS review logs (your own plus any shared decks).
 - Split: temporal split — train on earlier reviews, test on later ones, to mimic real studying and avoid leakage.
 - Metrics:
 - Log loss over held-out reviews per section and overall.
 - RMSE(bins): bin predicted recall probabilities (e.g., 0.0–0.1, 0.1–0.2, ...) and compute RMSE between predicted and empirical recall per bin.

- Brier score and decomposition (reliability, resolution, uncertainty) for transparency.
- Comparison:
 - Baseline FSRS-6 as configured in Anki (retention target consistent across sections).
 - Your adjusted model (e.g., topic-aware scheduling).
- Output:
 - Calibration plots per section (BBFL, CPBS, PSBB, CARS).
 - A short MCAT-specific note: “When we predict 0.8 recall on BBFL facts, actual recall is ~0.79–0.82 on held-out reviews.”

2. MCAT performance model evaluation (novel MCAT-style questions)

• Objective

- Predict correctness on new MCAT-style passage questions and quantify the gap between recall (memory) and performance (application), especially under interleaving vs. blocking.

• Design

- Build dataset:
 - For a subset of topics across BBFL, CPBS, PSBB, and CARS, create or collect MCAT-like passage questions (from practice sources or your own generation with traceable sources).
 - For each topic t , design ≥ 2 paraphrased questions that test the same idea in new words (paraphrase test requirement).
 - Log: FSRS recall probs $M_{s,t}$, interleaving indicators, spacing, practice testing counts, response times.
- Models:
 - Baseline: $P_q = f(M_{s,t})$ only (no learning-science variables).
 - Extended: $P_q = f(M_{s,t}, I_{s,t}, S_{s,t}, T_{s,t}, D_q)$.
- Evaluation:
 - Accuracy on held-out questions per section.
 - Log loss per section.
 - Calibration plots (predicted probability vs. actual correct fraction).
 - Analysis of “gaps”: cases where $M_{s,t}$ is high (e.g., ≥ 0.8) but P_q is low (e.g., ≤ 0.6), especially for paraphrased questions.

3. MCAT readiness mapping and give-up rule

- **Objective**

- Convert performance predictions into MCAT section scores (118–132) and total scores (472–528), with uncertainty ranges and abstention logic explicitly aligned to AAMC confidence bands.

- **Mapping plan**

- For each section s :
 - Sample or define a representative set of questions (weighted by MCAT content category weights) with difficulty parameters D_q .
 - Use your performance model to predict P_q for each question and simulate many “virtual test takers” to estimate a distribution of raw correct counts.
 - Apply an equating-style mapping (or a learned mapping from practice exam data if available) to convert raw scores to scaled scores $S_s \in [118, 132]$.
- Total score:
 - Sum the four section scaled scores to get $T = \sum S_s$, constrained to 472–528.

- **Confidence bands and give-up rule**

- Confidence bands:
 - Use simulation to compute a posterior distribution over S_s and T ; report bands similar to AAMC’s scheme (e.g., ± 1 for sections, ± 2 for total when uncertainty is high; narrower when you have lots of stable data).
- Give-up rule (example MCAT thresholds, you can tune):
 - At least $N_{\text{mem}} = 200$ graded reviews tagged to MCAT topics overall, with ≥ 50 per section.
 - At least $N_{\text{perf}} = 50\text{--}100$ graded MCAT-style questions per section (depending on data) with at least some representation of all major content categories.
 - Coverage $C \geq 50\%$: each section must have cards/questions covering $\geq 50\%$ of topics from the official or derived content outline.
 - If any threshold is not met, the app shows no readiness score, and explicitly lists missing data (“No MCAT score yet: only 38% BBFL topics covered; only 30 PSBB passage questions answered; minimum is 50.”).

4. MCAT interleaving feature ablation (learning-science study feature test)

- **Objective**

- Test a concrete MCAT-specific study feature based on interleaving and topic-aware ordering by comparing: 1. Full MCAT app (feature on). 2. MCAT app with interleaving feature off (blocked, same FRS engine). 3. Plain Anki MCAT deck.

- **Design**

- Pre-register:

- Study feature: MCAT interleaved sessions that mix topics within and across sections; topic-aware queue that prioritizes weak high-weight topics.
- Hypothesis: “Interleaving MCAT topics will raise accuracy on new mixed-topic, passage-based questions at the same study time compared with blocked practice and plain Anki.”
- Primary metric: percentage correct on a fixed set of mixed-topic MCAT exam questions after equal total study time across the three builds.
- Run:
 - Same group of learners, same MCAT deck, same question set, equal timed study sessions in all three conditions.
 - Log practice accuracy, test accuracy, and time.
- Analyze:
 - If full app > feature-off, the interleaving feature helps.
 - If full app > plain Anki, your whole MCAT app beats baseline.
 - Report effect sizes, ranges, and null results honestly (“Interleaving made no difference in CARS performance in this sample”) to align with Speedrun grading.

5. Leakage and MCAT AI card generation checks

• Objective

- Ensure MCAT test questions don’t leak into training data; ensure AI-generated MCAT cards are accurate and pedagogically useful before inclusion.

• Design

- Leakage script:
 - Compute TF-IDF or embedding vectors for all training cards and test questions; flag any training item whose similarity to a test item exceeds a threshold (e.g., cosine > 0.8).
 - Inspect flagged pairs, remove or adjust training items that are near-duplicates of test questions.
- AI card checker (MCAT):
 - Gold set: 50 Q/A pairs from trusted MCAT prep or AAMC materials, with known correct answers.
 - AI generation: create 50 cards from one MCAT source (e.g., a section of the content outline, a textbook chapter), using RAG with named sources.
 - Classification:
 - Correct & useful (accurate and tests a meaningful concept).
 - Wrong (incorrect fact or reasoning; automatically blocked).

- Correct but poor teaching (vague, trivial, duplicate).
- Pre-defined cutoff (e.g., $\geq 80\%$ correct & useful, 0% wrong) required before AI cards are allowed into the deck.

DOK 2: Knowledge Tree

How it's Structured and Supports the BrainLift:

- **Category:** Learning Science - Effective Study Techniques

• **Source:** Practice tests, spaced practice, and successive relearning: Tips for classroom use and for guiding students' learning (Dunlosky & Rawson, 2015)

DOK 1 - Facts:

- Practice testing and spaced practice are identified as high-utility techniques that consistently benefit learning across varying ages and abilities (Dunlosky & Rawson, 2015).
- Technique utility rankings: High (Practice testing, Spaced practice), Moderate (Elaborative interrogation, Self-explanation, Interleaved practice), and Low (Summarization, Highlighting, Keyword mnemonic, Imagery for text, Rereading) (Dunlosky & Rawson, 2015).
- Practice tests yield maximum benefits when they require retrieving answers from long-term memory (e.g., free recall) rather than simple recognition (Dunlosky & Rawson, 2015).
- Providing explanatory feedback (why an answer is correct or incorrect) during practice tests significantly enhances performance on transfer tests compared to providing corrective feedback alone (Dunlosky & Rawson, 2015).
- Successive relearning integrates practice testing with spaced intervals across multiple sessions to ensure information is reliably recalled (Dunlosky & Rawson, 2015).
- Spacing, interleaving, retrieval, generation, and varied practice each hurt short-term performance while strengthening long-term retention, a concept known as "desirable difficulties" (Bjork & Bjork, 2011).

DOK 2 - Summary:

- Students often rely on low-utility habits like cramming and highlighting, whereas high-utility, evidence-based strategies like spaced practice and practice testing offer superior, long-term learning outcomes (Dunlosky & Rawson, 2015; Bjork & Bjork, 2011).
- Practice tests function best when paired with delayed, explanatory feedback and when targeting key concepts, which directly improves students' ability to use and comprehend content on transfer tests (Dunlosky & Rawson, 2015).

- Successive relearning acts as a structured, multi-session framework that capitalizes on both testing and spacing to push students toward deeper mastery of target content (Dunlosky & Rawson, 2015).
- Learners frequently mistake "ease of recall" for mastery, which leads them to prefer inefficient, lower-utility study strategies over more effective, evidence-based ones (Bjork & Bjork, 2011).

• **Category:** Learning Science - Principles of Memory and Encoding

- **Source:** Teaching the science of learning (Weinstein, Madan, & Sumeracki, 2018)

DOK 1 - Facts:

- Spaced practice shows clear benefits for retention intervals longer than a month, and the extent of storage strength improvement is inversely related to current retrieval strength (Weinstein et al., 2018).
- Cramming results in high retrieval strength but low storage strength, leading to rapid forgetting, whereas spacing allows retrieval strength to wane, increasing storage strength (Weinstein et al., 2018).
- Interleaving study and test opportunities, or 'test-potentiated learning,' is often superior to blocking study and test phases (Weinstein et al., 2018).
- Retrieval practice improves long-term retention even without overt output (covert retrieval), but balancing success rates with difficulty is crucial, as excessively easy retrieval can undermine learning (Weinstein et al., 2018).
- Concrete examples and visual illustrations are highly effective for learning complex, multi-step concepts, especially for learners with low prior knowledge, as they often convey information more concisely than text (Weinstein et al., 2018).

DOK 2 - Summary:

- To maximize long-term retention for complex, multi-year content like the MCAT, learners should prioritize spaced practice over cramming, as cramming creates high immediate retrieval strength at the expense of long-term storage strength (Weinstein et al., 2018).
- Effective study design should interleave related topics and alternate study with testing, rather than blocking these activities separately, to leverage test-potentiated learning (Weinstein et al., 2018).
- While retrieval practice is a powerful tool, it must be balanced; retrieval should be challenging enough to produce gains in storage strength, avoiding 'too easy' retrieval situations (Weinstein et al., 2018).
- Educators should utilize concrete, visual examples to scaffold understanding of abstract or complex processes, ensuring these examples are consistent with the concepts being taught to avoid misapplication (Weinstein et al., 2018).
- **Link to source:** [Provide a direct URL to the source material.]

MCAT-aligned DOK1 – Facts and references

- **Sections:** The MCAT includes four sections: Biological and Biochemical Foundations of Living Systems (BBFL), Chemical and Physical Foundations of Biological Systems (CPBS), Psychological, Social, and Biological Foundations of Behavior (PSBB), and Critical Analysis and Reasoning Skills (CARS).
- **Scoring:** Each section is scored from 118 to 132, centered at 125. The total score is the sum of the four section scores, ranging 472–528, centered at 500.
- **Equating:** The AAMC uses equating to convert raw correct counts to scaled scores, ensuring consistency across different test forms.
- **Confidence Bands:** MCAT total scores are reported with a ± 2 -point confidence band; section scores with ± 1 -point bands. A reported 501 suggests the true score lies between 499 and 503. When comparing applicants, overlapping confidence bands mean score differences are less meaningful—a key insight for readiness uncertainty.
- **Content Outline:** Exam content is organized by foundational concepts and categories. Coverage is defined as the "percentage of outline topics with at least one high-quality card and at least one exam-style question in your deck".
- **Interleaving & Induction:** Kornell & Bjork (2008) found that interleaving artists' paintings improved inductive recognition, even though learners incorrectly believed blocking was better. Birnbaum et al. (2012) showed that interleaving's benefit relies largely on "discriminative contrast".
- **Math & Reasoning Analog:** Taylor & Rohrer (2010) demonstrated that interleaved math practice roughly doubled test performance while reducing "discrimination errors," providing a strong analog for MCAT reasoning practice.
- **High-Utility Techniques:** Spaced practice and practice testing are "high utility" techniques, yielding large effects on delayed tests ($d > 0.5$ – 0.6).
- **Memory Models (FSRS):** FSRS-6 significantly improves calibration over SM-2. Benchmark data (~350M reviews) shows FSRS-6 log loss ≈ 0.345 – 0.344 , RMSE(bins) ≈ 0.063 – 0.065 , and AUC ≈ 0.705 – 0.708 .
- **Readiness Metrics:** Log loss is the primary calibration metric. RMSE(bins) measures error between predicted and empirical recall. The Brier score (reliability, resolution, uncertainty) is used for probability predictions.
- **Readiness UI:** Readiness UI should mirror AAMC confidence bands, e.g., "Projected MCAT 508; likely range 503–512; confidence low because only 42% topics covered".

MCAT-aligned DOK2 – Relationships and design implications

- **Readiness Space:** The MCAT's 118–132 per-section scale and 472–528 total range define the output space. The model must predict section scores first, then sum to a total, respecting AAMC equating logic.
- **Passage-Based Reasoning:** MCAT sections (especially BBFL, PSBB, CARS) require passage-based application, not just discrete fact memory; performance on novel passage questions is the critical bridge.
- **Interleaving Application:** Interleaving MCAT topics (e.g., endocrine vs. nervous system) fosters inductive and discriminative learning, essential for recognizing which concept applies to a novel passage.
- **Memory Integration:** Spacing and practice testing modulate long-term retention. The memory model (e.g., FSRS) provides the base recall probability, M_s , while interleaving ($I_{s,t}$), spacing ($S_{s,t}$), and practice testing ($T_{s,t}$) act as modifiers.
- **Performance Prediction:** The model predicts probability $P_q = f(M_{s,t}, I_{s,t}, S_{s,t}, T_{s,t}, D_q)$, where D_q represents the question difficulty parameter akin to IRT item difficulty.
- **Uncertainty & Abstention:** AAMC's confidence bands (± 1 section/ ± 2 total) provide a native uncertainty scale. Predicted score ranges should reflect this—widening when data is sparse and narrowing as evidence increases.
- **Give-up Rule:** Because MCAT scoring is noisy, the system must practice abstention. Readiness scores should be withheld if coverage is low (e.g., <50% of topics), passage practice is insufficient (especially for CARS), or graded review volume is below threshold.